



## Bayesian characterization of the young open cluster NGC 6383 using HDBSCAN and Gaia DR3

L.M. Escobar-Pulgar<sup>1</sup> & N.A. Henríquez-Salgado<sup>1</sup>

<sup>1</sup> *Departamento de Astronomía, Facultad Ciencias Físicas y Matemáticas, Universidad de Concepción, Av. Esteban Iturra s/n Barrio Universitario, Casilla 160-C, Concepción, Chile*

**Abstract** / This study focuses on determinate the characteristics of the young open cluster NGC 6383. To achieve this, the HDBSCAN clustering algorithm is utilized to identify potential cluster members based on galactic coordinates, proper motions, and parallaxes sourced from Gaia DR3. We asses diverse parameters such as the tidal, cluster and core radius, distance throug parallax and isochrone-fitting, proper motion, age, metallicity, and others pertinent parameters of NGC 6383 using an extension to Hamiltonian Monte Carlo, the No-U-Turn Sampler. The results of this analysis point to NGC 6383 being a very young open cluster. Furthermore, our investigation reveals that the historical central binary HD 159176 is not a member of the cluster, leading to a revision of its age. The findings of this study contribute to the understading of NGC 6383 by providing more accurate and detailed characterization.

**Keywords** / open clusters and associations: individual — galaxies: star clusters: general — stars: distances — techniques: photometric — parallaxes — proper motions

**Contact** / lescobar2019@udec.cl